Cpts 317. 8/24.

Big picture:

Software Engg:

Software Design vs. Requirements,

language of all observable behaviors of the software system

alg design

Alg vs. Problem

language of all string encodings of all positive instances (Cpts 350)

⋮

317.

Automata vs languages.

technical path:

a. Hierachy of automata vs
" " languges.

(simple automata, simple languges,
Complex " , complex " ....).

b. So, we can understand
simple software, simple design,
simple alg, complex software,
Complex design, ...
simple problem, simple requments, ...

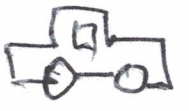
We start with the simplest class of languges, called regular languges.

Let Σ be a nonempty and finite set;

Σ is called alphabet where each element is called a symbol,

Exaple. $\Sigma = \{a, b\}$ has two symbols a, b.

$\Sigma = \{G, Y, R\}$ has three symbols. (green, yellow, Red)

$\Sigma = \{$,  $\}$.

Σ^k: the set of all words on alphabet Σ with length k;

Σ^2 with $\Sigma = \{a, b\}$

$= \{aa, bb, ab, ba\}$

$\Sigma^1 = \{a, b\}$

$\Sigma^0 = \{\Lambda\}$

empty word (with length 0).

Remark. $\Sigma^0 = \{\Lambda\} \neq \emptyset$.

Σ^* : the set of all words on alphabet Σ;

$$\Sigma^* = \bigcup_{k \geq 0} \Sigma^k$$

$$= \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \ldots\ldots$$

$\Sigma = \{a\}$. // unary alphabet.

$\Sigma^* = \{\Lambda, a, aa, aaa, \ldots\ldots\}$

$\Sigma = \{a, b\}$

$\Sigma^* = \{\Lambda, a, b, ab, ba, aa, bb,$
$aaa, \ldots\ldots\ldots\ldots\}$

---

all bit-strings with length 8:

$\{0, 1\}^8$

Let Σ be an alphabet. A language $\mathcal{L}$ is a subset of Σ^*; $L \subseteq \Sigma^*$.

not necc. proper

Let $\Sigma = \{a, b\}$. Example languages:

$L = \{a, aa\}$.

$L = \{a, aa, aaa, \ldots\}$

$L = \{ab, aabb, aaabbb, \ldots\}$

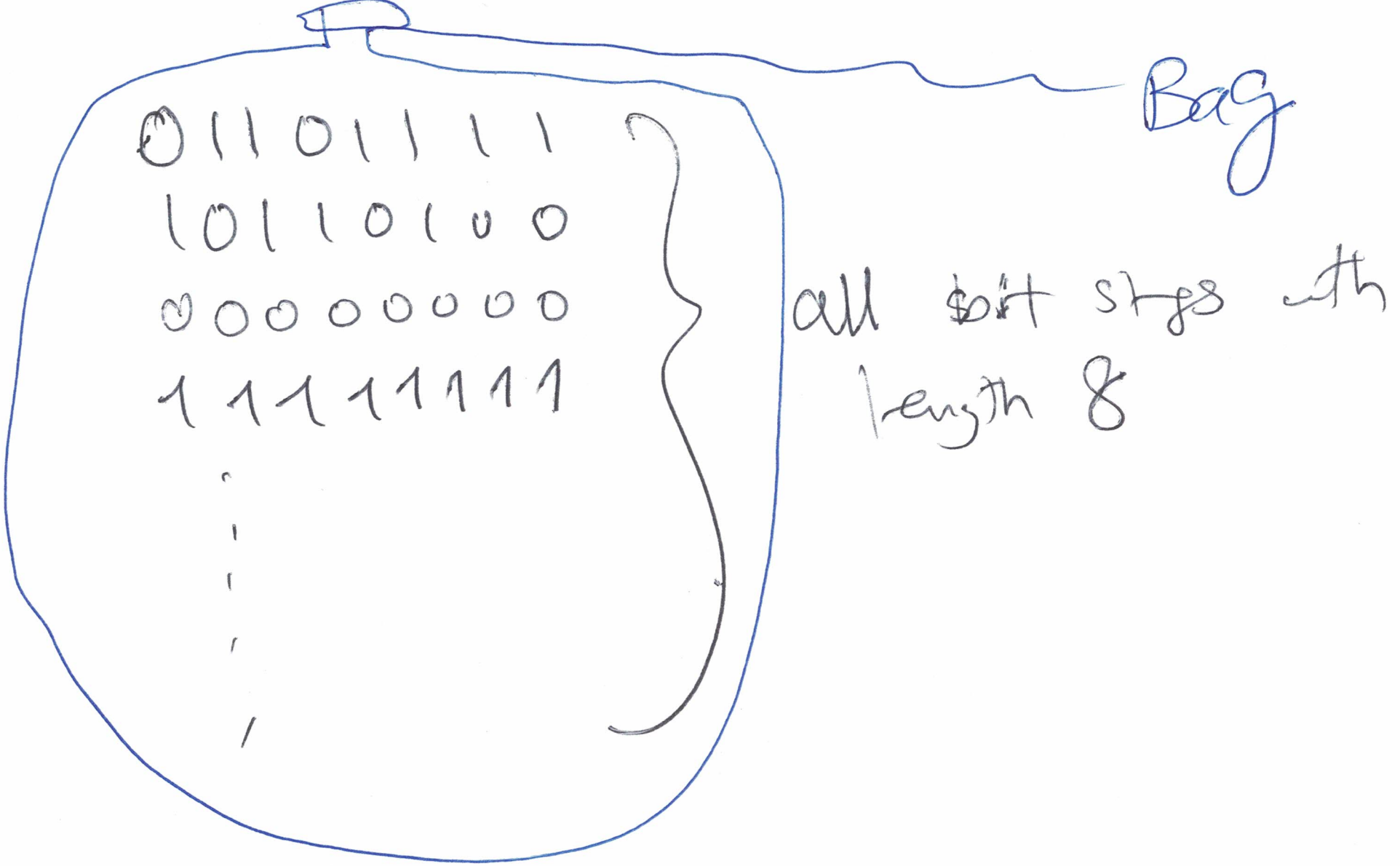
$\vdots$

Remark: language is a set of words,

NOT a word.

Remark: "A language is quite complex"

$\neq$ "A word in the language is quite complex."


Bag
01101111
10110100
00000000
11111111
all bit strings with length 8

$\omega_1 = abbaaabbaaabb$

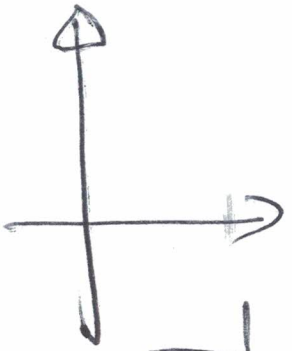
$\omega_2 = aaabbbbbbbbaaa$

Which one is more complex? why?

What do you mean by "a word is complex"?

Google it out!

a b b a b a



automaton is a ~~scanner~~ scanner from left to right

In this way, the automaton doesn't need to store the word always.